

Week 5 – Pipeline Automation with Azure DevOps

STEP 1 — Open Azure DevOps

Open:

[Azure DevOps](#)

Login using Microsoft account.

What Happens Here?

Azure DevOps dashboard opens.

This platform is used to:

- store project code
 - create pipelines
 - automate execution
 - manage repositories
 - generate reports automatically
-

STEP 2 — Create New Project

Click:

New Project

Project Details

Field	Example Value
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Project Name	CourseEnrollmentTracker
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Visibility Private

Then click:

Create

What Happens Internally?

Azure DevOps creates:

- project workspace
- repository
- pipeline environment
- execution environment

This project stores all online course tracking files.

STEP 3 — Create Python Script

Create file:

`weekly_report.py`

Python Script Code

```
import pandas as pd

# Load progress data
df = pd.read_csv("student_progress.csv")

# Filter students with less than 50% progress
low_progress = df[df['progress'] < 50]

# Save report
low_progress.to_csv("weekly_progress_report.csv", index=False)

print("Loading Student Progress Data")
print("Filtering Students Below 50% Progress")
print("Generating Weekly Report")
```

```
print("Saving Report")
print("Pipeline Completed Successfully")
```

Explanation of Each Line

Code	Meaning
Load progress data	Reads student progress dataset
Filter students below 50%	Finds slow learners/dropout risk
Save report	Generates weekly CSV report
Pipeline Completed Successfully	Execution completed

Why We Create This File?

This Python script automates:

- student progress checking
- low progress detection
- weekly report generation

Azure DevOps pipeline will automatically execute this script every week.

STEP 4 — Create requirements.txt

Create file:

`requirements.txt`

Add Libraries

pandas
numpy

Why requirements.txt is Important?

This file contains required Python libraries.

Before executing the project, Azure DevOps:

- reads this file
 - installs dependencies automatically
-

What Happens Without requirements.txt?

Pipeline may fail because required packages are missing.

Example error:

ModuleNotFoundError

STEP 5 — Create Sample CSV File

Create file:

`student_progress.csv`

Sample CSV Data

```
student_id,student_name,course,progress
101,Rahul,Python,75
102,Priya,SQL,40
103,Arun,PySpark,30
104,Meena,Azure,85
105,Karan,DevOps,45
```

Why This File?

This dataset contains:

- student names
- enrolled courses
- progress percentage

The Python script reads this data to generate reports.

STEP 6 — Upload Files into Repository

Go to:

Repos

Upload:

- weekly_report.py
 - requirements.txt
 - student_progress.csv
-

What is a Repository?

A repository stores project files.

Examples:

- GitHub repository
 - GitLab repository
 - Azure Repos
-

Why Repository is Needed?

Pipeline reads files directly from repository.

Without repository:

- pipeline cannot access files
- automation cannot happen

STEP 7 — Create Pipeline

Go to:

Pipelines

Click:

Create Pipeline

What Happens Here?

Azure DevOps opens pipeline configuration setup.

This pipeline defines:

- what to execute
 - how to execute
 - when to execute
-

STEP 8 — Select Repository

Choose:

Azure Repos Git

Then select repository containing:

- weekly_report.py
 - requirements.txt
 - student_progress.csv
-

Why This Step?

Pipeline must know where project files are stored.

STEP 9 — Configure Pipeline

Choose:

Starter Pipeline

Azure DevOps opens YAML editor.

What is YAML?

YAML is a configuration language.

It defines pipeline instructions.

Pipeline reads YAML and performs actions automatically.

STEP 10 — Replace Default YAML Code

Delete default code.

Paste:

trigger:

- main

pool:

vmImage: 'ubuntu-latest'

steps:

- script: |

pip install -r requirements.txt

python weekly_report.py

displayName: 'Run Weekly Student Progress Report'

YAML Explanation

trigger

trigger:

- main

Meaning:

Whenever code is pushed into the main branch,
pipeline runs automatically.

pool

pool:

vmImage: 'ubuntu-latest'

Meaning:

Azure DevOps creates temporary Ubuntu virtual machine for execution.

steps

Defines execution steps.

Install Dependencies

pip install -r requirements.txt

Meaning:

Install required Python libraries:

- pandas
 - numpy
-

Execute Python Script

python weekly_report.py

Meaning:

Runs weekly student progress analysis automatically.

displayName

displayName: 'Run Weekly Student Progress Report'

Meaning:

Step name shown in pipeline execution logs.

STEP 11 — Save and Run Pipeline

Click:

Save and Run

What Happens Internally?

Azure DevOps creates virtual machine

↓

Downloads repository files

↓

Reads YAML configuration

↓

Installs dependencies

↓

Runs Python script

↓

Generates weekly report

↓

Captures execution logs

↓

Shows pipeline status

STEP 12 — View Execution Logs

Expected logs:

Loading Student Progress Data
Filtering Students Below 50% Progress
Generating Weekly Report
Saving Report
Pipeline Completed Successfully

What are Execution Logs?

Execution logs show:

- what pipeline executed
 - whether execution succeeded
 - whether errors occurred
 - generated report status
-

Successful Pipeline Status

Pipeline status becomes:

Succeeded

Meaning:

- pipeline executed successfully
- weekly report generated
- automation completed successfully